

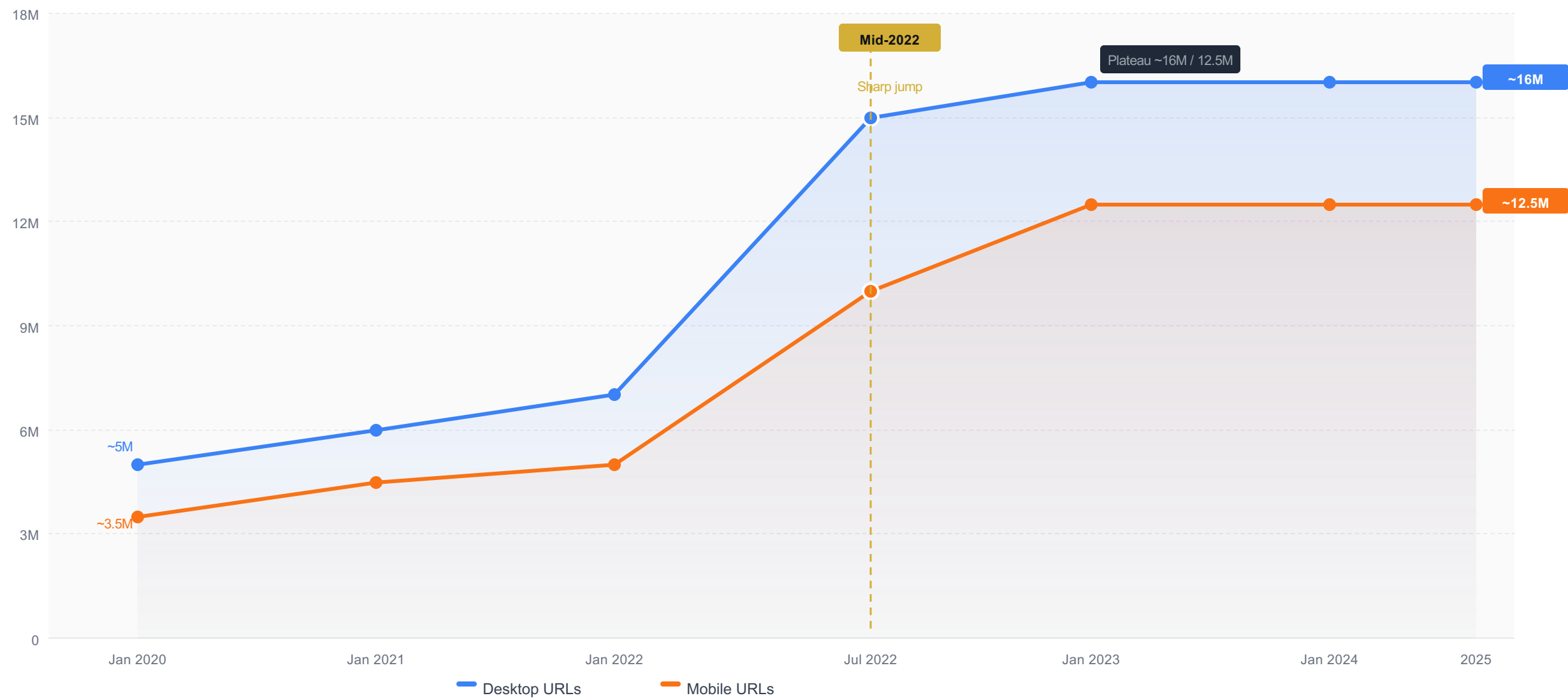
State of the Web 2020–2025

HTTP Archive Trends in Page Weight,
Protocols & Security

FOR WEB PERFORMANCE ENGINEERS & FRONT-END ARCHITECTS

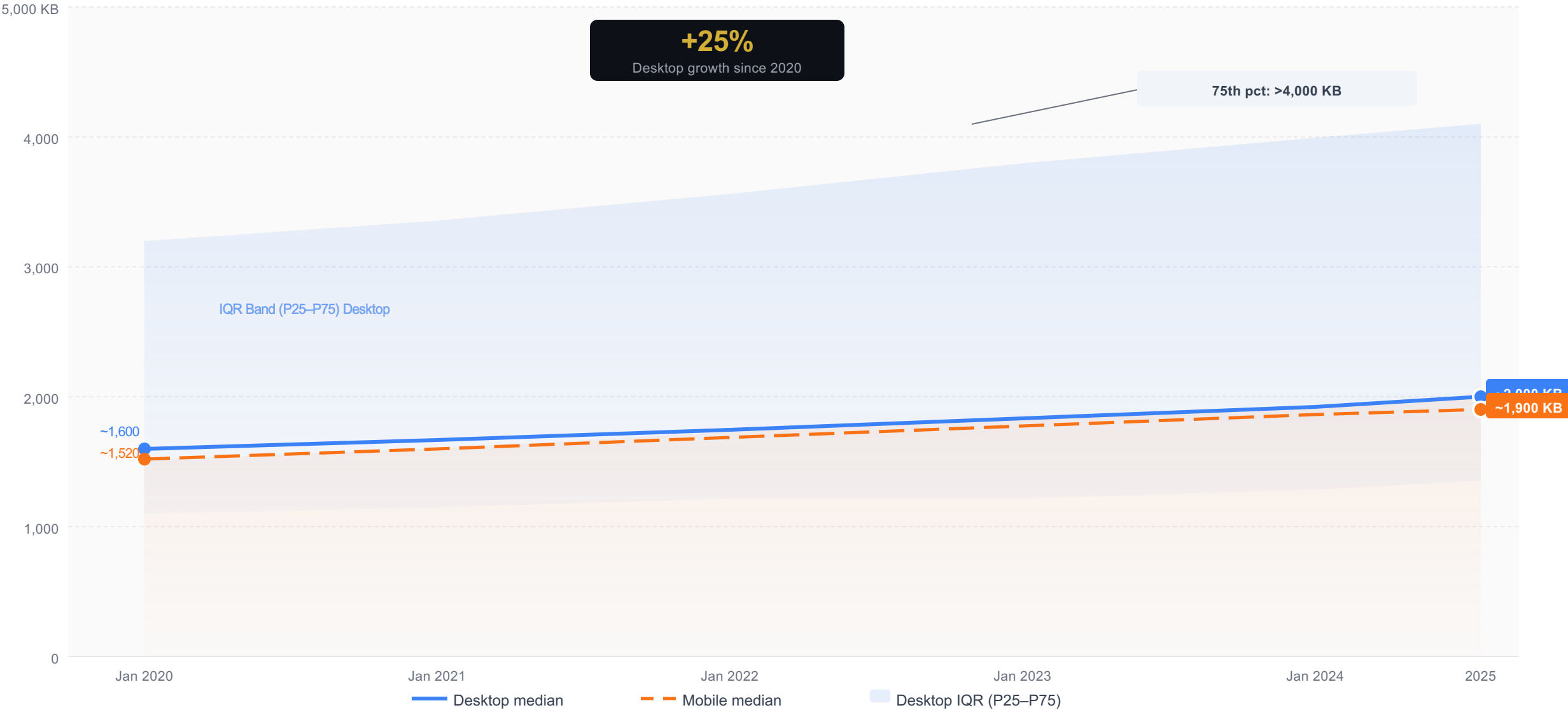
Crawl Sample Size Tripled: From ~5M to 16M Desktop URLs Since 2020

HTTP Archive URL crawl sample — Desktop and Mobile (Jan 2020 – mid-2025)



Median Page Weight Climbs Steadily to ~2,000 KB on Desktop (+25% Since 2020)

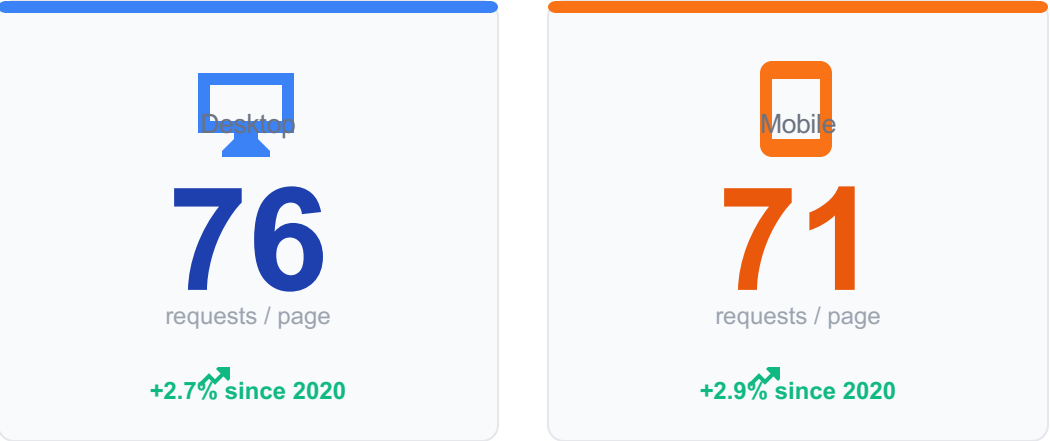
Total transfer size (KB) — Median with IQR bands · Desktop vs Mobile · Jan 2020 — 2025



76 Requests per Desktop Page — While TCP Connections Drop to 12

Requests per page (median) vs. TCP connections per page (median) · Desktop and Mobile

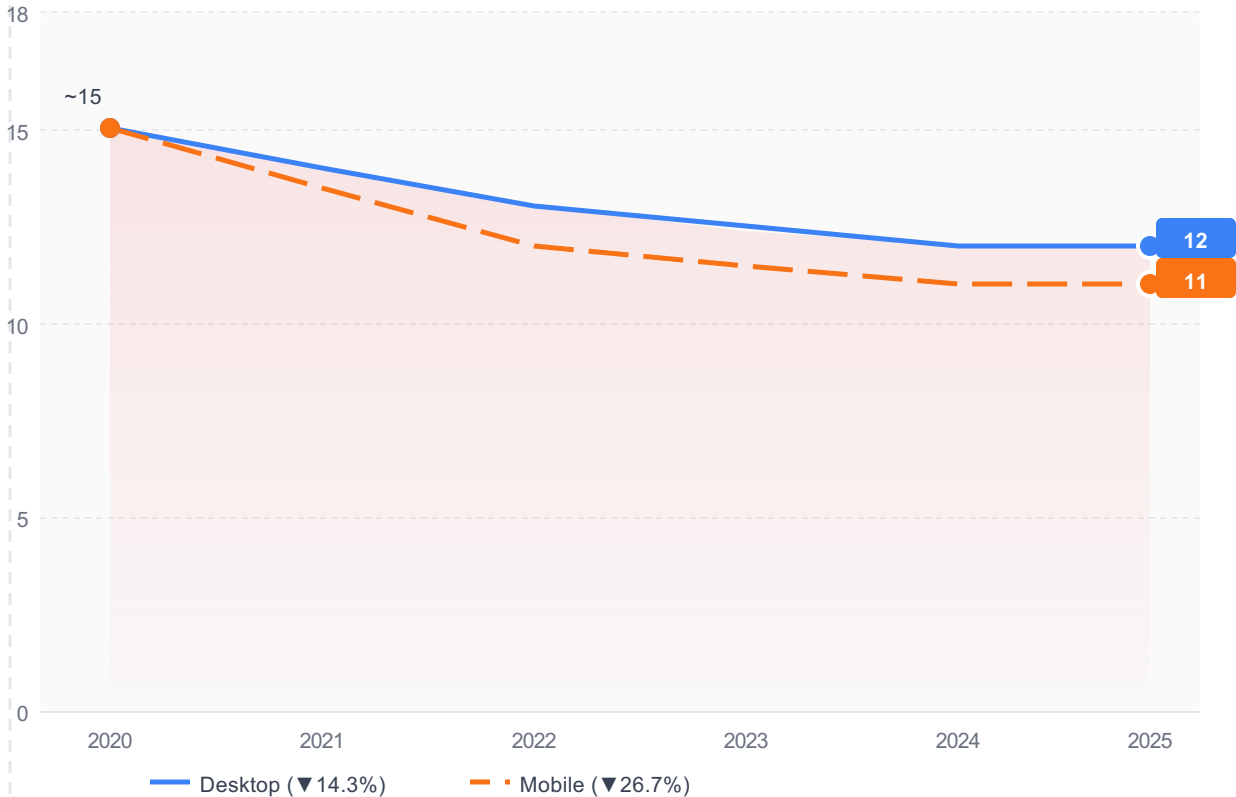
REQUESTS PER PAGE (MEDIAN)



Modest growth in resource requests

Request counts have grown only ~3% over 5 years despite pages becoming ~25% heavier — assets are consolidating, not multiplying.

TCP CONNECTIONS PER PAGE (MEDIAN)



Multiplexing drives the decline

HTTP/2 and HTTP/3 allow many requests over fewer TCP connections — improving network efficiency.

HTTPS Requests Reach 99%: Encryption Is Now the Default

Share of requests served over HTTPS · Desktop and Mobile · Jan 2020 – 2025

99%

of all web requests use HTTPS



Desktop

99.2%

▲ +19 pp since 2020



Mobile

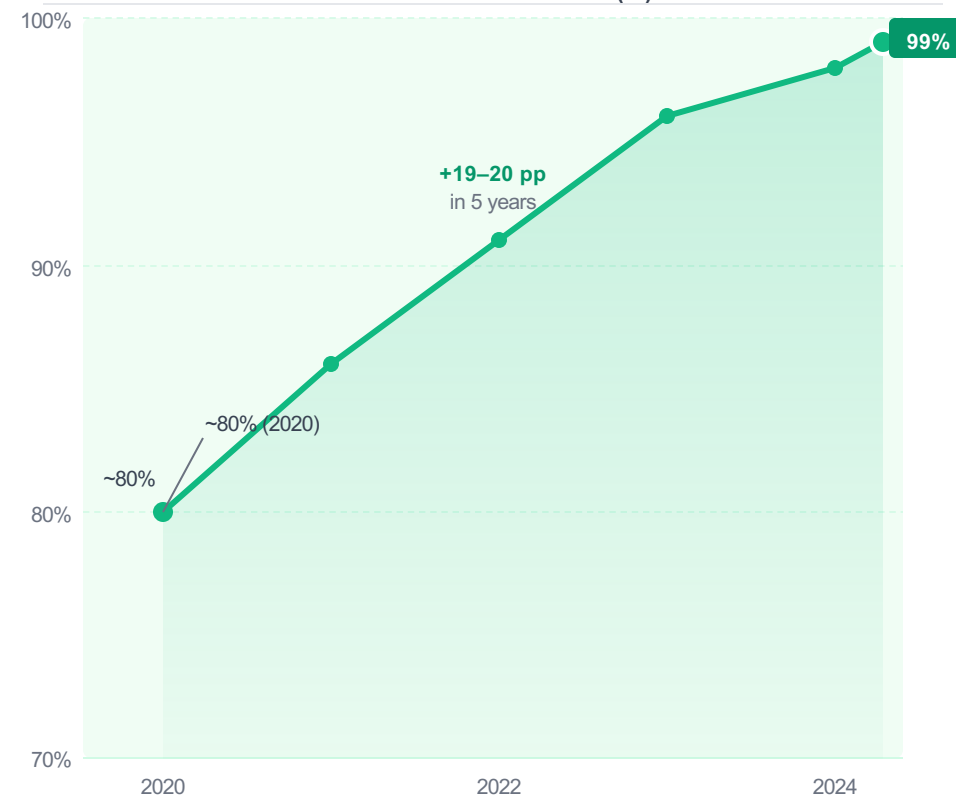
99.1%

▲ +20 pp since 2020



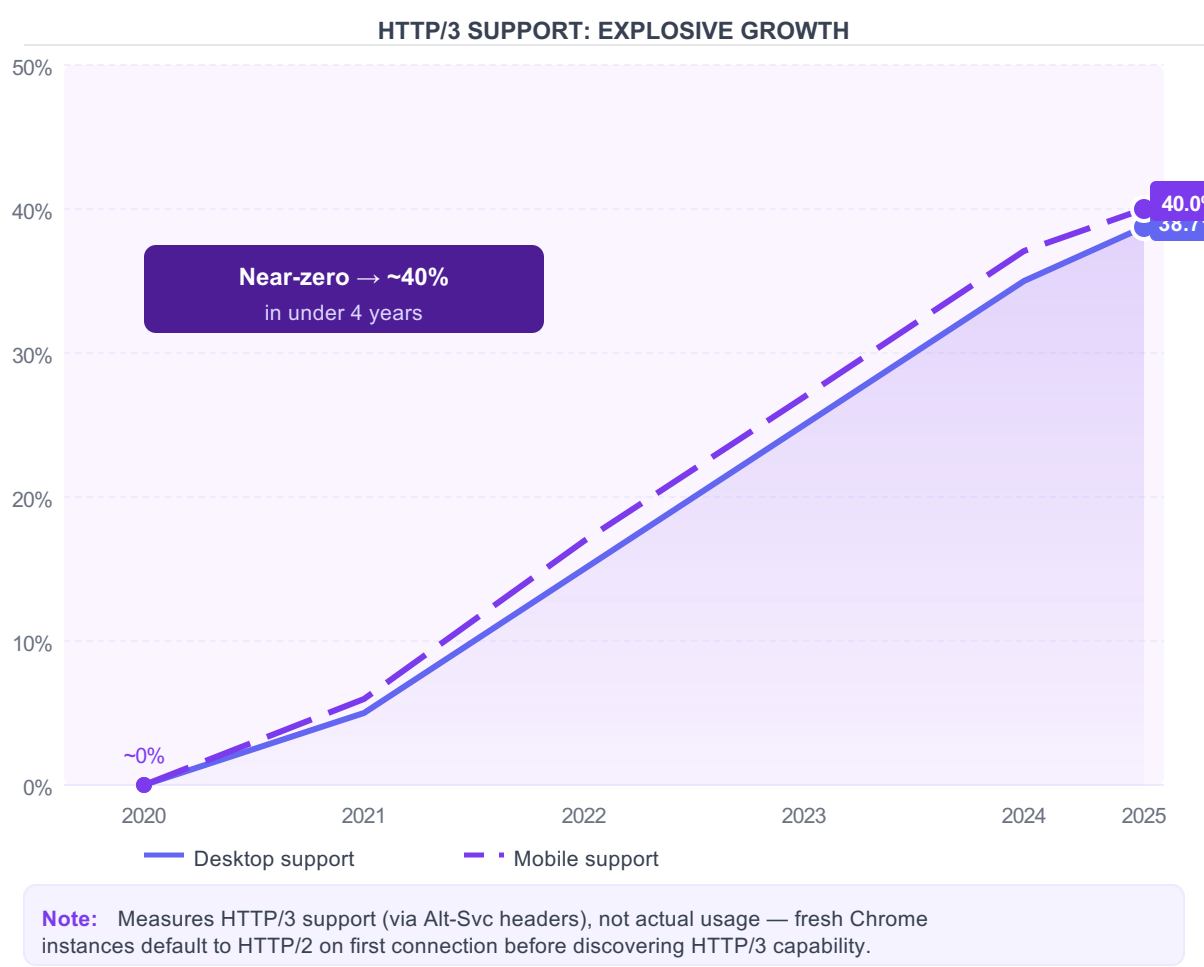
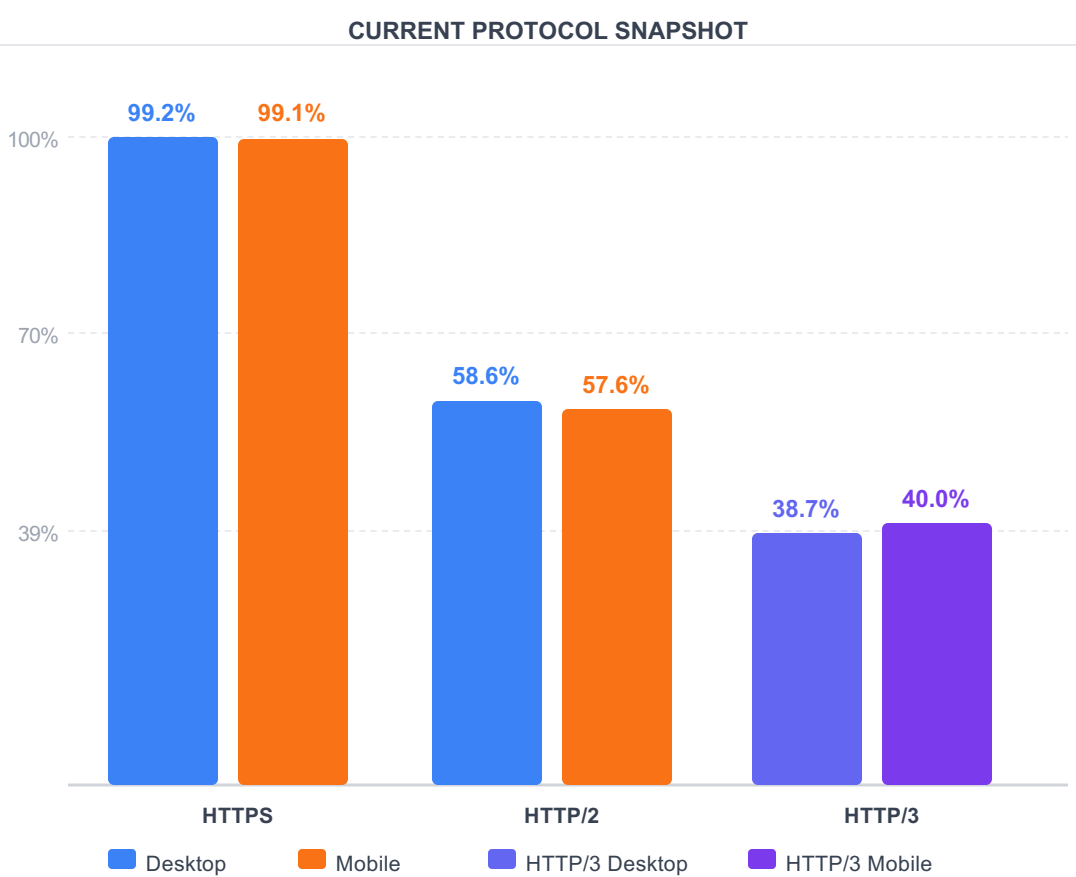
Plaintext HTTP is effectively extinct on the modern web

HTTPS Share Over Time (%)



HTTP/2 Holds at ~58% While HTTP/3 Support Surges to ~40%

Protocol adoption snapshot (current) · HTTP/3 growth trajectory from near-zero · Desktop and Mobile



Font-Display Usage Reaches 42% on Mobile — Still Room to Grow

Lighthouse audit: font-display CSS property adoption · Desktop 40.1% · Mobile 42.5% · vs. 2020 baseline

What is font-display?

The CSS **font-display** property controls how a web font is displayed while loading. Without it, browsers may show invisible text (FOIT) until the font loads.

⚠️ **FOIT = Flash of Invisible Text — a common UX degradation**

⚠️ More than half of pages still risk FOIT

With ~40–42% adoption, ~58% of pages do NOT use font-display, leaving users exposed to invisible text during font loading events.

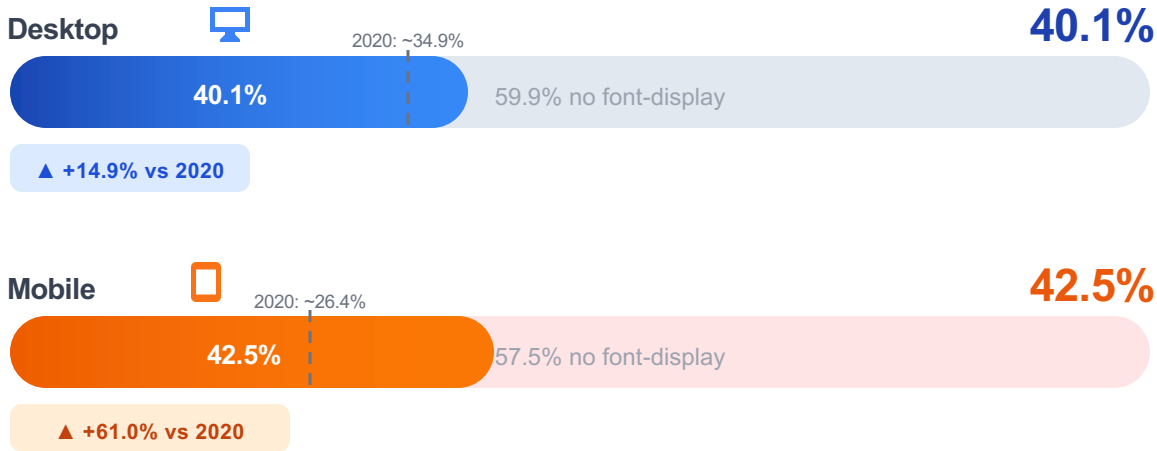
📈 Strong growth trajectory

Desktop adoption has grown **+14.9%** since 2020
Mobile adoption has grown **+61.0%** since 2020

💡 Opportunity:

Adding font-display: swap; is a single-line CSS change with immediate UX benefit.

FONT-DISPLAY ADOPTION RATE



Adoption vs. No-Adoption Breakdown



Key Takeaways: Bigger Pages, Smarter Protocols, Stronger Security



Page Weight Keeps Climbing

+25%

Median desktop pages now ~2,000 KB
— ▲ 25% since 2020, driven by richer media and JavaScript growth.



HTTPS Is the New Baseline

99%+

99%+ of requests are now encrypted.
Plaintext HTTP is effectively extinct on the modern web.



HTTP/3 Fastest-Growing Metric

~40%

From near-zero to ~40% support in under 4 years — rapidly becoming mainstream infrastructure.



Fewer TCP Connections

▼27%

Despite more requests, median TCP connections ▼15–27% since 2020.
HTTP/2 and HTTP/3 multiplexing wins.



Font-Display Still Underused

42%

Only ~40–42% of pages use font-display.
58%+ still risk Flash of Invisible Text (FOIT). A one-line CSS fix available.



Benchmark Your Site

Compare your own page weight, protocol adoption, and font-display coverage against these HTTP Archive baselines at httparchive.org/reports/state-of-the-web — updated monthly with fresh crawl data.

Metric	Desktop (Current)	Mobile (Current)	Desktop Δ	Mobile Δ
Total Transfer Size (median)	~2,000 KB	~1,900 KB	▲ ~25%	▲ ~25%
Total Requests (median)	76	71	▲ 2.7%	▲ 2.9%
TCP Connections / Page (median)	12.0	11.0	▼ 14.3%	▼ 26.7%
HTTPS Requests	99.2%	99.1%	▲ ~19 pp	▲ ~20 pp
HTTP/2 · HTTP/3 Support	58.6% · 38.7%	57.6% · 40.0%	Modest · Explosive	Modest · Explosive